Conflict, Climate, and Cash: Protecting Children in Nigeria's Conflict Zones[*]

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Abstract

This study examines the impact of armed conflict exposure on children's food insecurity in Nigeria's drier regions and assesses the protective role of cash transfers. Combining household survey data (MICS 2021) with geo-coded conflict records (UCDP 2021) and satellite climate data, we focus on areas with below-average soil moisture where specific armed groups are active. We employ an instrumental variable (IV) approach using current-year rainfall fluctuations – hypothesized to act as a tactical obstacle for ongoing conflict – to predict household exposure to conflict zones. Our findings indicate that living within 5km of a recent violent event significantly increases moderate food insecurity, raising the likelihood of skipping meals, eating less, consuming fewer kinds of food, or not eating healthy by 60-80 percentage points for households not receiving support. However, we find strong evidence that the N-Power cash transfer programme substantially mitigates these negative effects for recipient households in conflict zones. This protective effect is not observed for the most severe food insecurity indicators or for other cash transfer programmes like HUP or pensions. Robustness checks, including alternative conflict zone definitions, and placebo tests using lagged and nearby rainfall, support the validity of our IV strategy. The study highlights the severe food security consequences of conflict in vulnerable arid regions and underscores the potential, albeit context-specific, effectiveness of certain social protection designs in mitigating these impacts.

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1. Introduction

The negative impact of armed conflict on children and adolescents, particularly concerning food security, is well-documented; however, identifying effective protective interventions, especially in multi-crisis settings, remains a critical challenge. Nigeria, home to Africa's largest child population (UNICEF, 2023), starkly exemplifies this challenge. The country faces concurrent crises, including persistent internal conflicts like the Boko Haram insurgency, which overlap geographically with regions suffering significant climate vulnerability, particularly the arid conditions of the Northeast. This confluence is known to exacerbate childhood malnutrition and food insecurity in these zones (Dunn, 2018; Kaila & Azad, 2023).

While social protection programmes, especially cash transfers, are increasingly explored as potential buffers against conflict's effects (e.g., Ecker et al., 2024; Caeiro et al., 2025), evidence on their effectiveness in conflict zones is mixed, context-dependent, and often lacks robust causal identification (Premand & Rohner, 2024). Furthermore, conducting randomized controlled trials (RCTs) in conflict settings faces significant practical and ethical hurdles, requiring alternative approaches like quasi-experimental designs for rigorous impact evaluation (Puri et al., 2017; Verwimp, Justino, & Brück, 2019).

This paper contributes to understanding how to protect children in these complex environments by addressing key research gaps. Firstly, leveraging the established link between climate variability and conflict (e.g., Hsiang, Burke, & Miguel, 2013), we employ a novel instrumental variable (IV) approach using rainfall patterns to estimate the causal effect of conflict exposure on child food security in Nigeria. Secondly, we evaluate the effectiveness of three different social protection programmes in mitigating these negative impacts. This assessment focuses specifically on Nigeria's driest and most conflict-affected regions – areas facing a 'polycrisis' of conflict overlaid onto historical climate stress (Frimpong, 2020).

Our identification strategy hinges on the specific context of one-sided violence perpetrated by non-state actors, such as Boko Haram, in Nigeria’s most arid regions. We argue that within these particularly dry areas, short-term rainfall fluctuations are plausibly exogenous to household food security. Instead, such rainfall variations can serve as a natural experiment influencing the feasibility of ongoing conflict operations. This 'tactical obstacle' mechanism, previously documented in the literature (e.g., Raleigh & Kniveton, 2012; Sarsons, 2015), underpins our IV approach and is supported by our empirical findings, including robustness checks using lagged and nearby rainfall.

Using a cross-sectional dataset combining household surveys, geo-coded conflict records, and satellite-based climate data, our IV approach estimates that conflict exposure significantly exacerbates moderate food insecurity by approximately 60 to 80 percentage points for non-recipient households. In evaluating different social protection interventions, we find that Nigeria's N-Power conditional cash transfer programme appears effective in significantly mitigating this adverse impact of conflict, although its effect on more severe cases remains inconclusive. While our identification strategy leverages the climate-conflict nexus, our primary focus remains on evaluating social protection effectiveness within these challenging conflict zones historically affected by climate change.

This paper proceeds as follows: Section 2 provides context on Nigeria and reviews relevant literature. Section 3 details the data sources and measurement. Section 4 explains the empirical design and IV approach. Section 5 presents the results. Section 6 provides robustness checks, and Section 7 concludes with policy recommendations.

2. Background and Literature Review

Since its transition to civilian rule in 1999, Nigeria's internal conflict landscape has varied significantly by region. The Northeast has been the epicentre of Boko Haram's insurgency since 2010, a conflict resulting in extensive casualties (~20,000 deaths) and mass displacement (2.1 million people) over a decade (Kaila & Azad, 2023). Notably, as illustrated in Figure 1, the areas most affected by violent attacks in 2020 coincide with those experiencing the lowest soil moisture levels in Nigeria, particularly in the Northeast where Boko Haram operates. This geographical overlap highlights the confluence of conflict and environmental stress in the region. Consequentially, higher rates of childhood wasting and food insecurity have been documented in these conflict-affected zones (Dunn, 2018).

A growing body of evidence demonstrates the detrimental impact of armed conflict on food security within Nigeria and beyond. Using a multi-level model, Makinde et al. (2023) established a link between conflict exposure and increased likelihood of childhood stunting in Nigeria. Kaila & Azad (2023), employing panel fixed effects, similarly found that violence exposure in conflict zones exacerbates food insecurity. These findings resonate with research from other conflict contexts, including Ethiopia (Weldegiargis et al., 2023), Rwanda (Akresh, Verwimp, & Bundervoet, 2011), and the Gaza Strip (Brück, d'Errico & Pietrelli, 2019). The relationship can also be bidirectional, as food insecurity itself may trigger violence (Martin-Shields & Stojetz, 2019). Mechanisms driving this negative impact are varied, encompassing crop-land abandonment (Olsen et al., 2021), disruptions to agricultural production and markets, forced displacement, livelihood loss, destruction of food reserves, impeded aid delivery, and the psychological toll on affected populations (Shemyakina, 2022). Even the 'fear' of conflict can alter behaviour and negatively affect child health outcomes (Tapsoba, 2023).

Amidst the challenges posed by conflict, attention is turning towards identifying effective protective measures, particularly social protection programmes like cash transfers. Evidence on their effectiveness in conflict settings is accumulating. Studies demonstrate positive impacts: Ecker et al. (2024) showed cash transfers mitigating adverse nutritional effects in Yemen; Caeiro et al. (2025) used RCT and panel data from Ethiopia's Tigray conflict to show a "cash plus" programme lessened conflict's negative effect on food expenditure; and Tranchant et al. (2019) found food transfers reduced child food insecurity during conflict in Mali. This builds upon the established role of cash transfers in reducing food insecurity in sub-Saharan Africa more broadly (Burchi, Scarlato, & d'Agostino, 2018) and specifically during climate shocks like droughts (Banerjee et al., 2024). However, the relationship between cash transfers and conflict is not always positive; studies from Niger (Premand & Rohner, 2024), Indonesia (Sumarto, 2021), and Nepal (De Juan, Pierskalla, & Schwarz, 2020) suggest they can sometimes provoke conflict, whereas evidence from the Philippines indicates a conflict-reducing effect (Crost, Felter, & Johnston, 2016). This highlights that context and programme design are critical.

Building upon this literature, our paper makes several contributions. Firstly, we employ a robust causal identification strategy, using rainfall as an instrumental variable (IV), to estimate the effect of conflict on food insecurity, particularly among children. Secondly, our work aims to deepen the understanding of social protection effectiveness in contexts grappling with multiple crises – specifically, conflict overlaid onto areas historically affected by climate change, such as the desertic parts of Nigeria. While we do not examine the direct effect of rainfall shocks on food security, we analyse how cash transfers perform in these climate-vulnerable, conflict-affected settings. This focus on "polycrisis" environments is crucial. To our knowledge, only Kayaoglu et al. (2024) have recently explored similar terrain, finding in Sudan that while climate shocks alone weren't significant drivers of food insecurity in their context, conflict was, and cash transfers helped mitigate conflict's negative impact.

Our third contribution lies in identifying the specific conditions under which cash transfers are protective against conflict's effects on child food security. We provide evidence specifically for one-sided conflicts perpetrated by organized non-state actors (like Boko Haram) targeting civilians in desertic areas. Furthermore, we propose and test a novel theory: in these desertic environments, rainfall acts as a logistical obstacle, thereby reducing the likelihood of this specific type of conflict. The plausibility of rainfall being exogenous to food security outcomes in these specific parts of Nigeria strengthens our IV approach. Robustness checks using placebo rainfall measures (past years, nearby areas) support our findings.

It is important to contrast our findings with evidence concerning other conflict types, particularly resource-based conflicts involving pastoralist groups, which are also prevalent in parts of Nigeria. Studies like McGuirk and Nunn (2024) show that decreased rainfall significantly increases conflict risk among transhumant pastoralists across Africa (a one standard deviation decrease leads to a 24% increase in risk), and Sakketa et al. (2025) find similar patterns in Ethiopia (a 1mm rainfall decrease raises conflict risk). Similarly, Eberle et al. (2024) link temperature increases to farmer-herder conflict. These findings suggest that for conflicts driven by resource scarcity (water, pasture), lack of rainfall exacerbates tensions. Our study, however, focuses on a different conflict dynamic – one-sided violence by insurgents in desertic regions – where we find increased rainfall appears to hinder conflict, likely due to logistical challenges it poses for attackers. Thus, our evidence on the protective role of cash transfers is specific to mitigating the effects of this one-sided violence in desertic Nigeria, and we could not provide similar evidence for the distinct dynamics of two-sided pastoralist conflicts often found in central Nigeria. Understanding these differing relationships between climate variables and specific conflict types is crucial for designing effective interventions. In this study, we delve into these factors, maintaining a particular focus on children and adolescents.

3. Data

In this paper, we use a cross-sectional dataset combining three different sources: geo-coded household surveys on children's characteristics, conflict records based on the date and location of violent events, and satellite climate data. To understand the effect of conflict on children and adolescents, we restrict the observations to individuals who are 19 years old and younger. We extract child outcomes and household characteristics from the UNICEF Multiple Indicator Cluster Survey (MICS-6, 2021) from Nigeria, conflict records from the Uppsala Conflict Data Program (UCDP, 2021), and via Google Earth Engine we access soil moisture records from NASA (2020-2021) and rainfall data from the Climate Hazards Group at the University of California, Santa Barbara (2020-2021). All these datasets are nationally representative, covering the entire Nigerian territory. MICS has a complex survey design, and we ensured to include survey weights, strata, and primary sampling units (PSU) in all the calculations described in Tables 1 to 4 and Appendix Tables 1 to 7 below. Furthermore, only observations with complete data across all food insecurity indicators were included to maintain consistency in the analysis.

Using household-level geocodes, we estimated whether a child lives in a conflict zone in Nigeria. Conflict is defined based on the UCDP data for one-sided violence: the use of armed force by an organized actor against civilians resulting in at least 25 deaths per year. Conflict zones are defined as areas within either 5 km or 10 km of such a violent event that occurred during the last year. To protect the identity of research subjects, MICS geocodes are randomly displaced between 2 and 5 km. Therefore, the closest reliable estimate of conflict exposure we could calculate was within 5 km of a violent event, though we also present the 10 km cutoff for comparison.

Table 1 presents descriptive statistics and a balance check between individuals living in 'treated' (conflict zones) and 'untreated' (non-conflict zones) areas, specifically focusing on the drier regions of Nigeria (where the Soil Moisture Index is below the national mean). Eight indicators approximate the experience of food insecurity, based on self-reports from the head of household regarding conditions over the last month (e.g., worried about food, skipped meals, went a day without eating due to lack of resources). These household-level reports suggest widespread vulnerability, with national averages indicating that around 70% of children and adolescents live in households experiencing some form of food insecurity. It is through these household reports that we gauge the potential impact of conflict on the food security environment children are exposed to.

The balance table (Table 1) shows that for both the 5km and 10km definitions of conflict zones, most household-reported food insecurity indicators are relatively balanced between treated and untreated groups, showing no statistically significant differences. Similarly, households receiving N-Power or Pension cash transfers appear balanced across the groups. However, significant imbalances exist for several key variables. Notably, children in conflict zones tend to be older, live in rural areas, belong to households with a higher wealth index, and are predominantly Muslim. Furthermore, environmental conditions like soil moisture (in both 2020 and 2021) differ significantly between conflict and non-conflict zones. Interestingly, average rainfall in 2020 showed no significant difference between groups using the 10km definition (p=0.795) and only a marginal difference at 5km (p=0.042), while 2021 rainfall was significantly lower in conflict zones. Given the observed imbalances in key socioeconomic and demographic characteristics, we will use the variation in current (2021) average rainfall as an instrumental variable (IV) in subsequent analyses. This approach leverages the quasi-random nature of rainfall fluctuations within these dry areas to help isolate the causal effect of conflict exposure from the underlying differences between groups.

Figures 1a, 1b, 2, and 3 illustrate the study's geographical context. Figure 1a displays the distribution of all violent conflict events across Nigeria against soil moisture patterns. Our analysis, however, concentrates on armed conflict within Nigeria's drier, more desertic regions. We select these areas because their specific characteristics, often linked to climate change impacts, have seen the proliferation of armed groups like Boko Haram. Figure 1b thus narrows the geographical scope to regions with below-average soil moisture (SMI < National Average) and maps only incidents of one-sided violence against civilians. This map highlights how the specific type of conflict under study concentrates in these arid zones. Figure 2 further examines these dry areas, plotting the spatial variation in 2021 average annual rainfall against the locations of one-sided violence. We propose that rainfall variability within these dry regions acts as a quasi-random factor potentially influencing the intensity or occurrence ('turning on and off') of ongoing conflict dynamics.

Restricting the analysis to these drier areas, Figure 3 displays the spatial correlation between food insecurity (using "ate few kinds of food" as an example indicator) and the locations of one-sided conflict events. A clearer association appears under these restrictions compared to the national overview, suggesting that this specific type of conflict in arid zones is more strongly linked to localized food insecurity. Notably, there is a visible lack of MICS survey observations in Borno state (northeast Nigeria), an area known for significant conflict clusters. This data gap stems from severe security constraints that prevented the MICS survey team from fully covering the area. While MICS employs sampling weights to ensure remaining observations are representative, this lack of coverage in a high-conflict region might lead to an underestimation of the true treatment effects of conflict on food insecurity in our analysis.

Table 1 provides details on the coverage of social protection programmes in Nigeria within the analysed sample (drier regions). N-Power supports unemployed youth aged 18-35 with monthly stipends of ₦30,000 (approximately $38 USD), vocational training, and public work opportunities, lasting 1–2 years (Isah et al., 2024). About 2-3% of children in this sample live in households benefiting from N-Power, with no significant difference between conflict and non-conflict zones. The Household Uplifting Programme (HUP), also known as the National Cash Transfer Programme, targets poor and vulnerable rural households, offering a ₦5,000 stipend (approximately $6.40 USD) and conditional top-ups of ₦5,000 for school attendance and health check-ups. It also includes capacity-building initiatives to improve livelihoods (NCTO, 2024). Significantly fewer children in conflict zones live in households benefiting from HUP (2%) compared to non-conflict zones (6-7%). Additionally, pensions provide monthly payments to retired formal employees aged 60 and above; about 3% of children live in households receiving pensions, with no significant difference between groups. These estimates consider only households exclusively receiving one of these programmes.

Finally, Table 1 includes a list of individual characteristics used as control variables in this study, highlighting the imbalances mentioned earlier. Across the sample in drier areas, the average age is around 8.5-9.8 years, roughly half are female, and the majority in conflict zones live in rural areas (74-77%) compared to non-conflict zones (29-32%). The average wealth index is significantly higher in conflict zones (around 3.6) compared to non-conflict zones (around 2.4). A very small percentage identify as Christian in conflict zones (2-3%) compared to non-conflict zones (15-16%), while the opposite is true for Muslim identification (97-98% vs 84-85%).

4. Empirical Design

To explore the causal effect of armed conflict on children's and adolescents' food insecurity, we employ an IV approach using rainfall fluctuations as an instrument to identify households located in conflict zones. This approach aligns with the use of rainfall as an IV to predict conflict, which is well established in the economics literature (e.g., Miguel, Satyanath, & Sergenti, 2004). However, our focus differs in that we use the IV not to isolate the causal effect of economic growth on conflict but to isolate the causal effect of conflict on food insecurity among children. Recent evidence suggests that rainfall can also predict conflict through mechanisms unrelated to income, such as its impact as a tactical obstacle (Sarsons, 2015).

This IV approach addresses the issue of endogeneity between conflict and food insecurity, where both factors may influence each other. As discussed in Section 2, there is evidence supporting both the impact of conflict on food insecurity and food insecurity's potential role as a trigger for conflict. To address this, we restrict our sample to households located in areas with an average soil moisture index below the national mean during the previous year. This restriction helps isolate the endogenous relationship between rainfall and food insecurity, as rainfall in prior years can directly impact food security through agricultural productivity. By limiting the sample to the driest areas of Nigeria, we mitigate this potential endogenous link and concentrate on regions where violence is most prevalent.

Further, we use current-year rainfall as an instrumental variable, arguing that it serves as a tactical or logistical obstacle to conflict. Rainfall variability in the present year is randomly distributed across these arid, high-conflict areas, providing an exogenous source of variation that allows us to more accurately estimate the causal impact of conflict on food insecurity. This focus on regions with low soil moisture also aligns with findings in political science, which indicate that arid areas with weak state presence create conditions where insurgent groups can assert control due to logistical obstacles, such as difficult terrain and sparse populations, that hinder state governance (Herbst, 2014). While Nigeria experiences various types of violence often linked to poverty and food insecurity, the violence in these conflict zones typically aligns with the strategic goals of insurgent groups, making it more likely to act as an exogenous shock to local communities.

Our first-stage equation models the likelihood of a household being located in a conflict zone as a function of the average annual rainfall levels in that location, along with a set of control variables.

$$Conflict\ Zone_i = \alpha_0 + \alpha_1 Rainfall_{it} + \alpha_2 \boldsymbol{X}_i + \varepsilon_i \tag{1}$$

$Conflict\ Zone_i$, the main treatment variable, is defined as household (i) being located within 5 kilometres of at least one violent event that occurred during the last 12 months (t) and that resulted in at least one fatality[1]. $Rainfall_{it}$, the instrument, corresponds to the average amount of rainfall during the same period in the household's location[2]. $\boldsymbol{X}_i$ corresponds to a vector of individual characteristics, including the children's age, sex (female = 1, male = 0), rural residence (1 = rural, 0 = urban), the household's wealth index (ranging from 1 to 5), and religious affiliation, with the main options being Muslim (1 = Muslim, 0 = otherwise) or Christian (1 = Christian, 0 = otherwise). This equation also includes an error term to account for unobservable factors.

The outcomes in the second-stage equation indicate whether a child or adolescent lives in a household reporting food insecurity. Additionally, we include a term to indicate whether the child lives in a household receiving social protection through one of three possible

[1] For anonymization purposes, MICS survey geocodes are randomly displaced between 2 and 5 kilometres (km). As a result, 5 km was the closest approximation to a conflict zone that we were able to incorporate into our model. Robustness checks were conducted using 10, 15, and 20 km radii.

[2] Rainfall in t was calculated as the average rainfall at the household level during the 12 months preceding the survey, collected between September and December 2021. Robustness checks were also conducted using rainfall in t−1 and t−2, but only rainfall during the same period effectively works as an IV to isolate the causal effect of conflict on food insecurity.

programmes: *N-Power cash transfers*, *HUP cash transfers*, or a *retirement pension*. The same control variables and an error term are also included.

$$Food\ Insecurity_i = \beta_0 + \beta_1 \widehat{Conflict}\,Zone_i + \beta_2 Social\ Protection_i + \beta_3 \left(\widehat{Conflict}\,Zone_i \times Social\ Protection_i\right) + \beta_4 \boldsymbol{X}_i + \mu_i \quad (2)$$

The interaction term in the second-stage equation allows us to explore potential heterogeneous effects of armed conflict on food insecurity based on social protection status. Specifically, the coefficient β_3 estimates how the impact of living in a conflict zone on food insecurity differs for individuals in households receiving social protection compared to those in households not receiving such support within these zones. This helps understand if social protection modifies the relationship between conflict exposure and food insecurity.

For a proper causal interpretation of the IV approach regarding the effect of conflict (β_1), two main assumptions must be met. First, the treatment must be correlated with the instrument (relevance), meaning there must be a correlation between conflict zones and rainfall fluctuations in Nigeria. Second, the instrument should only affect the outcome through the treatment variable (exclusion restriction), implying that rainfall should impact food insecurity solely through its effect on conflict within this context. The following section begins by providing evidence to support these two assumptions.

5. Results

Column 1 in Table 2 presents the first-stage results for our IV approach using the 5km conflict zone definition. It demonstrates that each 1 mm increase in average rainfall in 2021 is associated with a 0.3 percentage point reduction in the likelihood of a household being in a conflict zone affected by one-sided violence during the same year (significant at p<0.01). This finding supports the relevance condition required for the IV approach—a clear correlation between the instrument (rainfall) and the treatment (conflict zone exposure). This relationship is further validated by a strong first-stage F-statistic of 25.52, well above the conventional threshold, indicating that average rainfall is a strong instrument for predicting conflict likelihood in this context.

Turning to the second-stage results in Table 2, the coefficients in columns 2 to 4, representing the most severe food insecurity indicators (A Day Without Eating, Hungry and Did Not Eat, Ran Out of Food), are not statistically significant. This suggests we cannot draw firm conclusions about the effect of this type of conflict or the *N-Power* programme on these extreme outcomes within this sample.

However, when examining columns 5 to 8, which focus on more moderate indicators of food insecurity, we observe statistically significant effects. Column 5 shows that for households not receiving N-Power, living in a conflict zone increases the likelihood of skipping a meal by approximately 71.8 percentage points ($\beta 1=0.718$, $p<0.05$). The interaction term ($\beta 3=-2.510$, $p<0.05$) is negative and significant, indicating that the effect of conflict on skipping meals is significantly less severe for households receiving N-Power compared to those not receiving it in conflict zones. While the coefficient for receiving N-Power alone ($\beta 2=0.163$, $p<0.05$) is positive, potentially reflecting targeting of more vulnerable households, the large negative interaction suggests a strong mitigating heterogeneous effect of the program during conflict.

Similarly, in column 6 (Eating Less), conflict increases the likelihood by 58.7 percentage points for non-recipients ($\beta 1=0.587$, $p<0.10$), and the interaction term shows a significant mitigating effect for N-Power recipients ($\beta 3=-2.133$, $p<0.05$). For column 7 (Ate Few Kinds of Food), conflict increases the likelihood by 82.7 percentage points for non-recipients ($\beta 1=0.827$, $p<0.05$), with a significant mitigating interaction effect ($\beta 3=-2.543$, $p<0.05$). In column 8 (Not Eating Healthy), conflict increases the likelihood by 67.8 percentage points for non-recipients ($\beta 1=0.678$, $p<0.05$), again with a significant mitigating interaction effect ($\beta 3=-2.401$, $p<0.05$). The final column presented (Worried About Food) shows a similar pattern, though the effect of conflict itself is not significant (β1=0.368), the interaction term remains significant (β3=-2.038, $p<0.05$), suggesting N-Power helps alleviate worry specifically in conflict zones. The consistently large, negative, and significant interaction terms across these moderate indicators strongly suggest that the N-Power programme substantially buffers households against the negative impacts of conflict exposure on food security[3]. These key coefficients (β1 and β3) are graphically displayed for all food insecurity outcomes in Figure 4, which visually confirms the general pattern of conflict increasing food insecurity (positive coefficients for 'Conflict Zone') and N-Power mitigating this effect in conflict zones (negative coefficients for the interaction term).

In Table 3 and Table 4, we replicate the analysis using the *Household Uplifting Programme (HUP)* and *retirement pensions*, respectively. Across both tables, we find very few statistically significant effects. For HUP (Table 3), the only significant coefficient related to conflict is β1 in column 5 (Eating Less), suggesting conflict increases this outcome by 125.2 percentage points ($p<0.10$) for non-recipients, but the interaction term is not significant. For pensions (Table 4), none of the conflict or interaction coefficients reach statistical significance. This general lack of significant results suggests that, unlike N-Power, neither HUP nor pensions demonstrate a measurable mitigating effect on food insecurity within conflict-affected areas in this specific analytical framework, possibly due to differences in programme design, targeting, or benefit levels relative to the shock of conflict.

6. IV Robustness Checks

This section details checks performed to assess the validity and robustness of our IV approach and findings. Appendix Tables 1 through 4 replicate the main analysis for N-Power

[3] Given the use of a Linear Probability Model with binary outcomes, coefficients representing percentage point changes can sometimes exceed 1 or be below -1, particularly with strong interaction effects. They indicate the direction and statistical significance of substantial effects.

(from Table 2) using alternative buffer zones of 10km, 15km, 20km, and 25km, respectively, to define conflict zones. The results remain broadly consistent with the 5km findings for the 10km, 15km, and 20km buffers, showing significant adverse effects of conflict on moderate food insecurity for non-recipients and significant mitigating interaction effects for N-Power recipients. However, as shown in Appendix Table 4 (25km buffer), the coefficients generally lose statistical significance. This pattern suggests the impact of conflict exposure on food insecurity diminishes as distance from violent events increases, supporting the idea that our main findings capture a localized effect.

Appendix Tables 5 and 6 present placebo tests using lagged rainfall from one year prior (t-1) and two years prior (t-2) as alternative instruments. While lagged rainfall might correlate with conflict through agricultural or economic channels, these tables show it does not yield statistically significant estimates of the conflict effect on food insecurity in the second stage. Furthermore, the first-stage F-statistics are considerably weaker, particularly for the two-year lag (F=2.23). These results support our choice of current-year rainfall as the primary instrument. The fact that only current rainfall works effectively aligns with our argument that it acts primarily as a tactical or logistical obstacle for ongoing conflict in these dry regions, rather than affecting food security directly or through slower economic mechanisms, thus bolstering the exclusion restriction assumption.

Appendix Table 7 introduces another crucial placebo test using the average rainfall in nearby areas (within 5km of the household location, but not the location itself) as an instrument. The rationale is that if rainfall primarily affected food security directly (e.g., through agriculture), nearby rainfall might also capture this effect. However, the results show that nearby rainfall is a weak predictor of conflict in the first stage (F=7.86) and yields no significant effects in the second stage. This finding further strengthens the exogeneity assumption of our main instrument (local, current-year rainfall). It suggests that rainfall's impact in this specific context operates locally, consistent with the tactical obstacle mechanism affecting conflict, rather than through broader agricultural channels that might also be captured by nearby rainfall. Focusing on one-sided violence, prevalent in these dry areas and often driven by insurgent groups' strategic goals rather than purely economic desperation, also helps satisfy the exogeneity requirement.

7. Conclusion and Policy Recommendations

The contribution of this paper is twofold. First, focusing on one-sided violence within the driest regions of Nigeria, it provides robust IV estimates showing that living in a conflict zone significantly increases moderate food insecurity among children and adolescents, particularly affecting indicators such as "skipping meals," "eating less," "eating fewer kinds of food," and "not eating healthy." Second, the paper offers new insights into the potential role of social protection in conflict settings. We find evidence suggesting the N-Power cash transfer programme significantly mitigates these adverse effects of conflict exposure on moderate food insecurity indicators for recipient households. However, this mitigating effect was not apparent for the most severe indicators of food insecurity, nor was it found for the HUP or retirement pension programmes.

The lack of significant results for extreme food insecurity might stem from N-Power's design, which doesn't primarily target the absolute poorest, or potentially from sample attrition where the most vulnerable are displaced or deceased. The absence of effects for HUP and pensions could relate to lower benefit amounts (especially HUP) compared to N-Power or differences in targeting that may not align as well with conflict-affected populations in this sample.

Our findings also underscore the interplay between climate conditions, conflict, and food insecurity. The IV strategy relies on rainfall fluctuations within Nigeria's driest areas—regions vulnerable to climate variability where specific armed groups have proliferated. The robustness checks, particularly the failure of lagged and nearby rainfall as effective instruments, support the interpretation of current rainfall as a valid instrument working through a tactical/logistical channel on ongoing conflict, exogenous to food security in this specific context. This highlights the value of programmes like N-Power in potentially addressing the compounded vulnerabilities arising from conflict and climate challenges in arid regions.

For policymakers, these results suggest that cash transfer programmes, particularly those with characteristics similar to N-Power (e.g., potentially higher benefit level), can be a valuable tool to buffer households against some of the food security consequences of conflict exposure in vulnerable, dry regions. While not a panacea for the complex impacts of violence, such programmes appear to alleviate significant hardship. Future research should delve deeper into the specific features of N-Power that drive its apparent effectiveness in this context and explore potential synergies with peacebuilding or climate adaptation initiatives. Investigating whether such transfers could, in turn, influence conflict dynamics themselves in these climate-stressed regions remains a critical avenue for inquiry.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work, the authors used ChatGPT-4 to assist with proofreading. Following its use, the authors reviewed and edited the content as necessary and take full responsibility for the final version of the publication.

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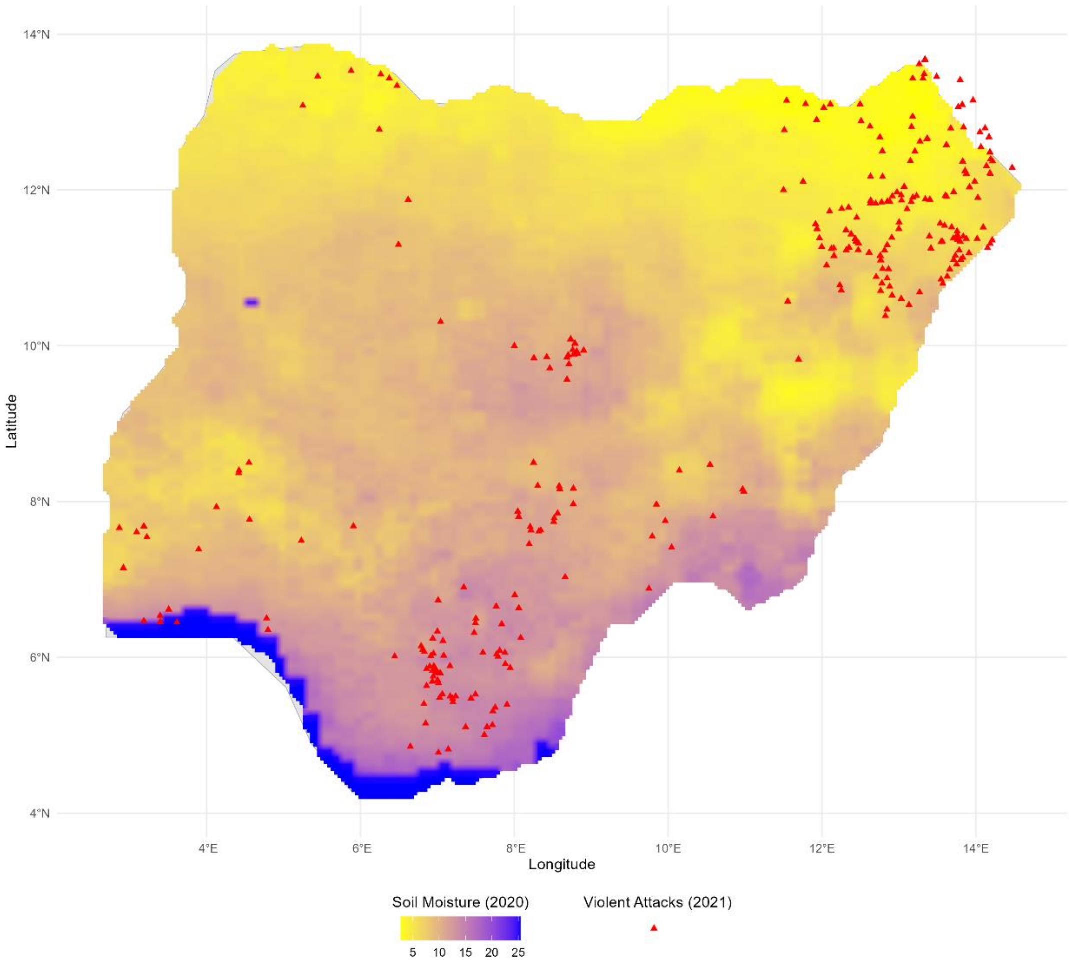
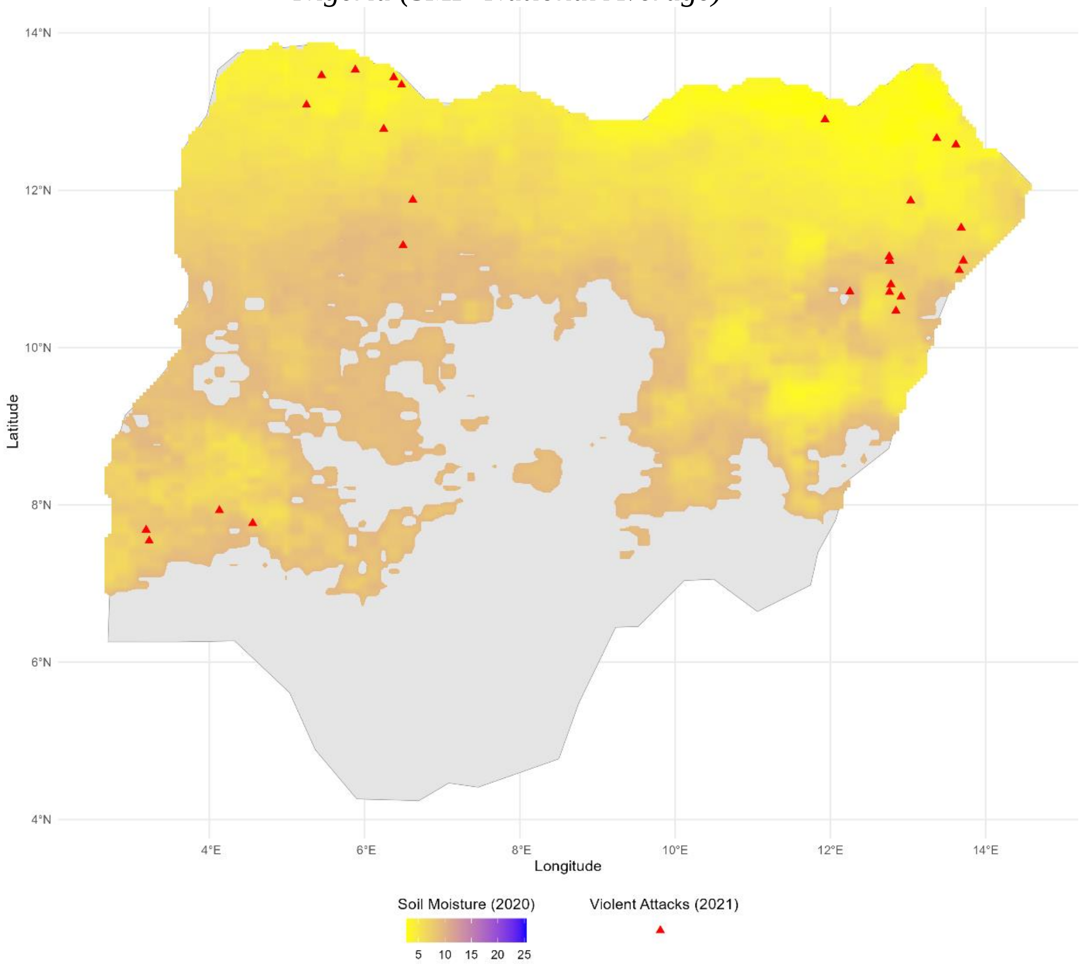
FIGURES

Figure 1a. *Map of Conflict and Average Soil Moisture Index in Nigeria*



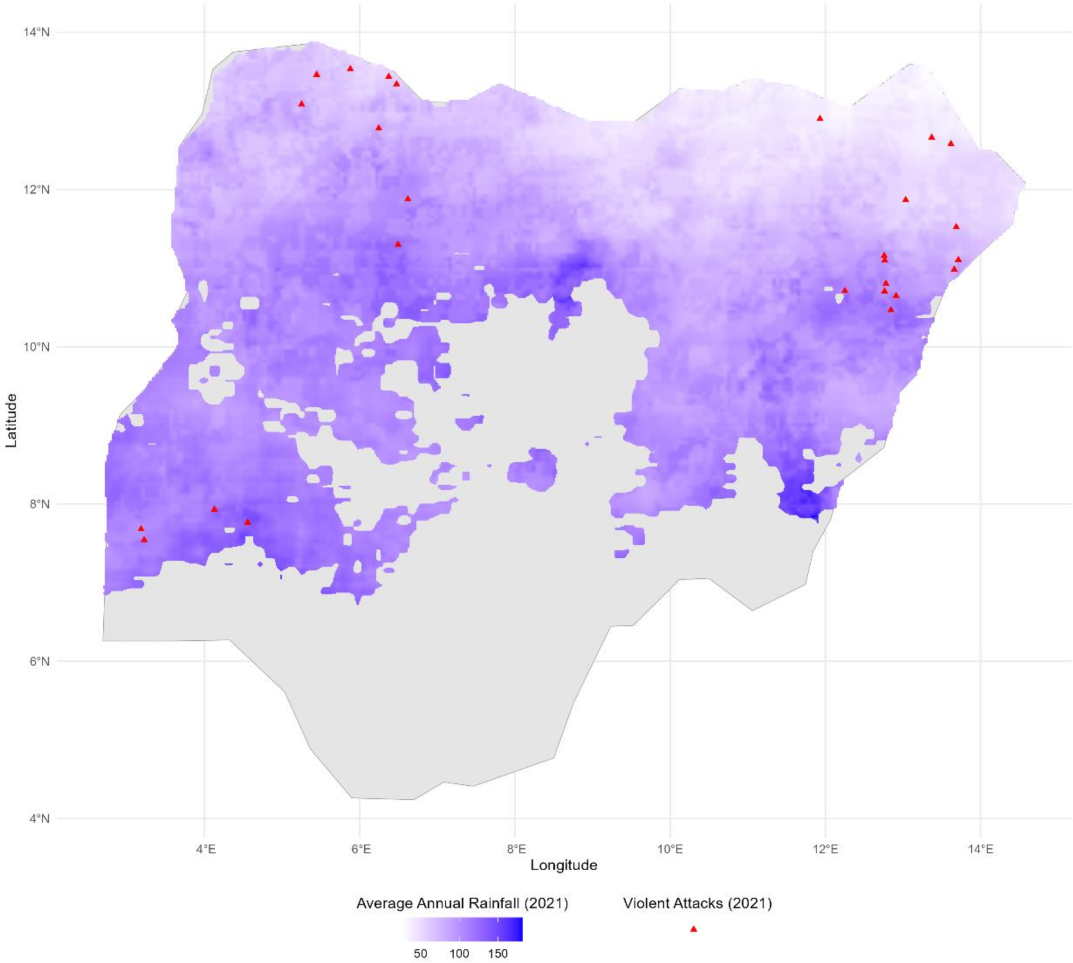
Note: Own calculations using UPPSALA Conflict Data Program (UCDP, 2021) and satellite soil moisture data from NASA (2020).

Figure 1b. *Map of One-Sided Conflict and Average Soil Moisture Index in Dry Areas of Nigeria (SMI<National Average)*



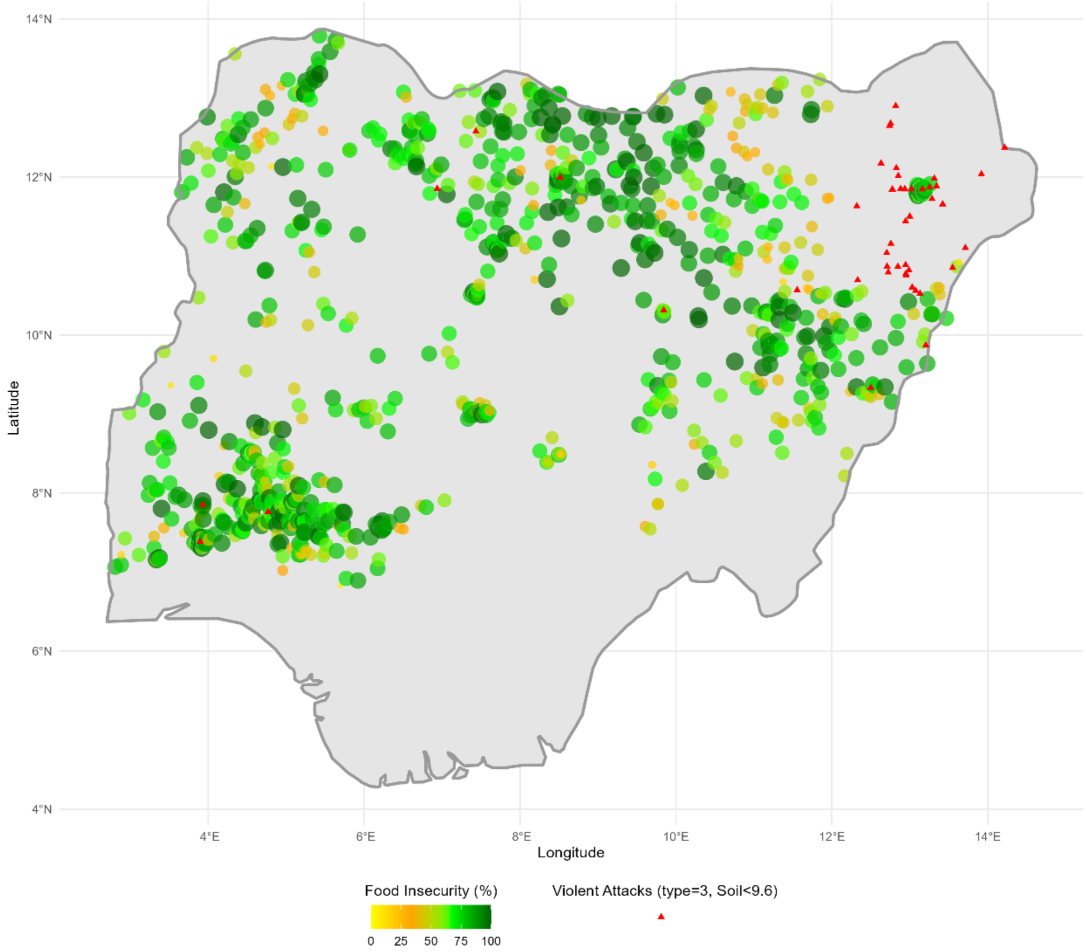
Note: Own calculations using UPPSALA Conflict Data Program (UCDP, 2021) and satellite soil moisture data from NASA (2020).

Figure 2. *Map of One-Sided Conflict and Annual Rainfall in Dry Areas of Nigeria (SMI<National Average)*



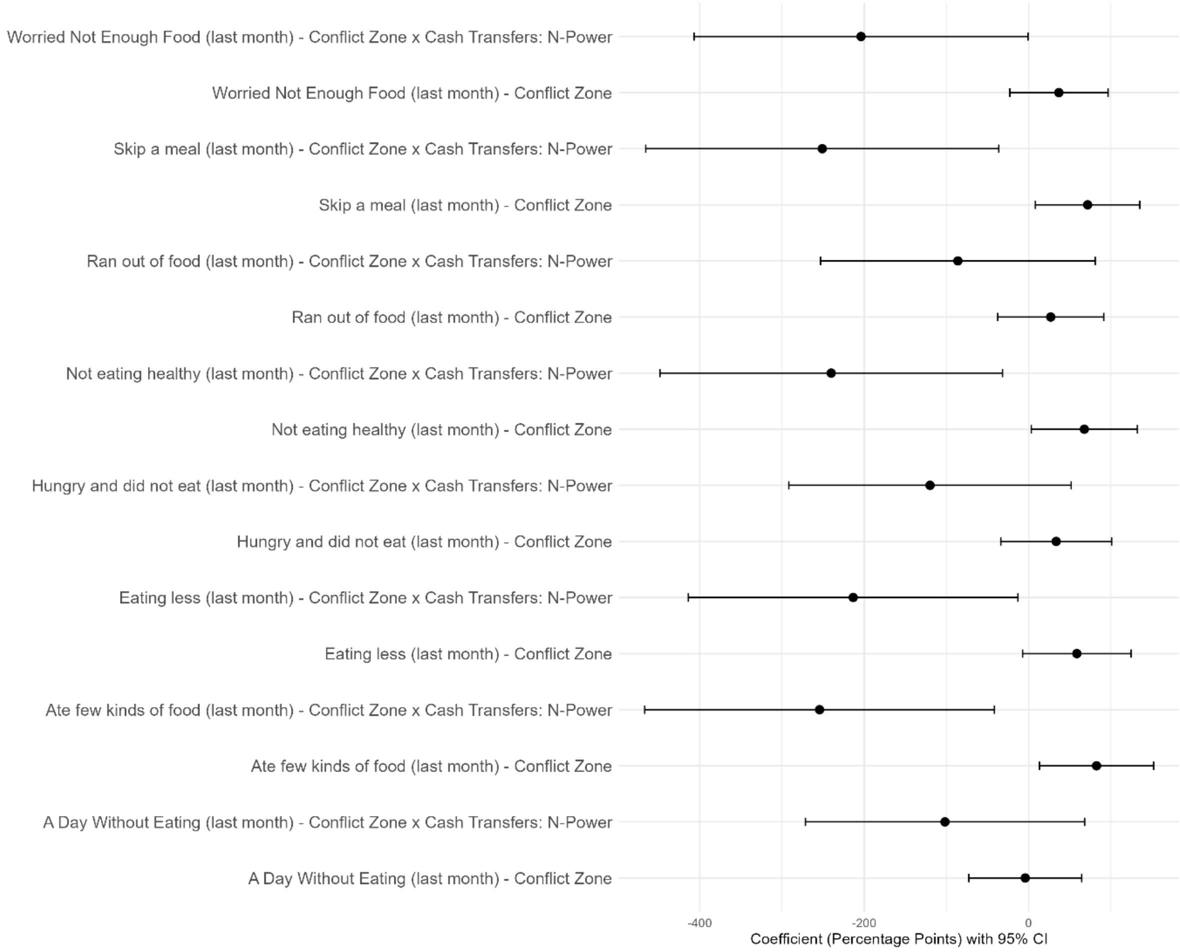
Note: Own calculations using UPPSALA Conflict Data Program (UCDP, 2021) and annual rainfall data from the Climate Hazards Group at UCSB (2021).

Figure 3. *Map of Conflict and Food Insecurity in Dry Areas of Nigeria (SMI<National Average)*



Note: Food insecurity indicator: Ate few kinds of food. Own calculations using UPPSALA Conflict Data Program (UCDP, 2021) and household survey data from UNICEF MICS (2021).

Figure 4. *Estimated Effects of Conflict on Food Insecurity Indicators*



Note: Coefficients and standard errors extracted from Table 2.

TABLES

Table 1. *Balance and Descriptive Statistics by Conflict Zone in Dry Parts of Nigeria (Soil Moisture Index<National Mean)*

	Conflict Zone: 5Km								**Conflict Zone: 10 Km**							
	Yes			**No**					**Yes**			**No**				
	Mean	SD	N	Mean	SD	N	Diff	p_value	Mean	SD	N	Mean	SD	N	Diff	p_value
	(1)	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**	**(7)**	**(8)**	**(1)**	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**	**(7)**	**(8)**
Food Insecurity: Worried (last month)	0.72	0.45	2,460	0.71	0.45	49,693	0.01	0.862	0.70	0.46	4,044	0.72	0.45	48,109	-0.02	0.492
Food Insecurity: Not eating healthy (last month)	0.75	0.43	2,441	0.73	0.44	48,719	0.02	0.564	0.73	0.44	4,074	0.73	0.44	47,086	0.00	0.897
Food Insecurity: Ate few kinds of food (last montl	0.74	0.44	2,428	0.72	0.45	48,383	0.02	0.544	0.73	0.44	3,961	0.72	0.45	46,850	0.01	0.823
Food Insecurity: Skip a meal (last month)	0.75	0.43	2,088	0.73	0.45	42,245	0.02	0.534	0.77	0.42	3,306	0.73	0.45	41,027	0.04	0.182
Food Insecurity: Eating less (last month)	0.75	0.43	2,330	0.72	0.45	44,995	0.03	0.349	0.76	0.43	3,754	0.71	0.45	43,571	0.05	0.086
Food Insecurity: Ran out of food (last month)	0.69	0.46	1,981	0.68	0.46	38,109	0.01	0.854	0.73	0.44	3,213	0.68	0.47	36,877	0.05	0.135
Food Insecurity: Hungry and did not eat (last mon	0.75	0.43	1,661	0.69	0.46	33,707	0.06	0.082	0.74	0.44	2,646	0.69	0.46	32,722	0.05	0.129
Food Insecurity: A Day Without Eating (last mont	0.70	0.46	1,026	0.71	0.46	22,046	0.00	0.966	0.68	0.47	1,577	0.71	0.45	21,495	-0.02	0.537
Average Soil Moisture Index (2020)	6.03	1.09	3,171	6.90	1.65	62,595	-0.86	0.000	5.96	0.86	5,185	6.96	1.67	60,581	-1.00	0.000
Average Soil Moisture Index (2021)	5.00	1.67	3,171	6.16	2.08	62,595	-1.17	0.001	4.45	1.40	5,185	6.30	2.06	60,581	-1.85	0.000
Average Rainfall (2020)	108.64	21.12	3,171	118.56	22.30	62,595	-9.92	0.042	117.31	21.33	5,185	118.16	22.46	60,581	-0.86	0.795
Average Rainfall (2021)	85.85	18.40	3,171	108.50	21.10	62,595	-22.66	0.000	88.52	14.48	5,185	109.60	21.14	60,581	-21.08	0.000
Cash Transfers: N-Power	0.03	0.16	1,837	0.02	0.14	32,639	0.01	0.539	0.02	0.15	2,931	0.02	0.14	31,545	0.00	0.831
Cash Transfers: HUP	0.02	0.15	1,137	0.06	0.24	22,175	-0.04	0.002	0.02	0.13	1,807	0.07	0.25	21,505	-0.05	0.000
Cash Transfers: Pension	0.03	0.18	1,883	0.03	0.16	31,753	0.01	0.548	0.03	0.17	3,110	0.03	0.16	30,526	0.00	0.938
Age of Respondent	9.64	5.59	3,171	8.58	5.44	62,595	1.06	0.000	9.75	5.50	5,185	8.50	5.43	60,581	1.25	0.000
Gender (Female=1; Male=0)	0.52	0.50	3,171	0.49	0.50	62,595	0.03	0.009	0.50	0.50	5,185	0.49	0.50	60,581	0.01	0.320
Rural Location (1=Yes; 0=No)	0.74	0.44	3,171	0.32	0.47	62,595	0.42	0.000	0.77	0.42	5,185	0.29	0.45	60,581	0.48	0.000
Wealth Index Quintile	3.63	0.94	3,171	2.43	1.30	62,595	1.20	0.000	3.58	1.04	5,185	2.36	1.28	60,581	1.22	0.000
Christian Religion (1=Yes; 0=No)	0.03	0.18	3,171	0.15	0.36	62,595	-0.12	0.000	0.02	0.14	5,185	0.16	0.36	60,581	-0.14	0.000
Muslim Religion (1=Yes; 0=No)	0.97	0.18	3,171	0.85	0.36	62,595	0.12	0.000	0.98	0.15	5,185	0.84	0.37	60,581	0.14	0.000

Note: Household survey data are taken from UNICEF MICS (2021), conflict data from the UPPSALA Conflict Data Program (UCDP, 2021), rainfall data from the Climate Hazards Group at UCSB (2020-2021), and soil moisture data from NASA (2020-2021).

Table 2. *Heterogeneous Effects of Conflict on Child Food Insecurity in Nigeria: N-Power Cash Transfers*

	Conflict Zone (1st Stage)	A Day Without Eating (2nd Stage)	Hungry and Did Not Eat (2nd Stage)	Ran Out of Food (2nd Stage)	Skip a Meal (2nd Stage)	Eating Less (2nd Stage)	Ate Few Kinds of Food (2nd Stage)	Not Eating Healthy (2nd Stage)	Worried About Food (2nd Stage)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
IV: Average Rainfall (2021)	−0.003***								
	(0.001)								
Conflict Zone (5Km)		−0.041	0.335	0.269	0.718**	0.587*	0.827**	0.678**	0.368
		(0.350)	(0.344)	(0.329)	(0.324)	(0.337)	(0.354)	(0.329)	(0.305)
Cash Transfers (N-Power)		0.021	−0.029	−0.113	0.163**	0.084	0.194**	0.158**	0.078
		(0.115)	(0.120)	(0.130)	(0.073)	(0.096)	(0.092)	(0.072)	(0.084)
Conflict Zone x Cash Transfers		−1.017	−1.199	−0.860	−2.510**	−2.133**	−2.543**	−2.401**	−2.038**
		(0.866)	(0.876)	(0.852)	(1.095)	(1.023)	(1.085)	(1.063)	(1.036)
Num.Obs.	9,168	9,168	9,168	9,168	9,168	9,168	9,168	9,168	9,168
1st Stage F-Statistic	25.52								

*Note: The sample consists of children and adolescents (aged <19) living in the driest areas of Nigeria (SMI < national mean). All coefficients are calculated using 2SLS and including the following control variables: Age, Female, Rural, Wealth Index, and Religion. Household survey data are taken from UNICEF MICS (2021), conflict data from the UPPSALA Conflict Data Program (UCDP, 2021), rainfall data from the Climate Hazards Group at UCSB (2020-2021), and on soil moisture data from NASA (2020-2021). *p<0.1; **p<0.05; ***p<0.01*

Table 3. *Heterogeneous Effects of Conflict on Child Food Insecurity in Nigeria: Household Uplifting Programme (HUP)*

	Conflict Zone (1st Stage)	A Day Without Eating (2nd Stage)	Hungry and Did Not Eat (2nd Stage)	Ran Out of Food (2nd Stage)	Skip a Meal (2nd Stage)	Eating Less (2nd Stage)	Ate Few Kinds of Food (2nd Stage)	Not Eating Healthy (2nd Stage)	Worried About Food (2nd Stage)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
IV: Average Rainfall (2021)	−0.002***								
	(0.001)								
Conflict Zone (5Km)		0.249	0.654	0.893	0.786	1.252*	0.904	0.691	0.574
		(0.649)	(0.673)	(0.679)	(0.564)	(0.735)	(0.617)	(0.546)	(0.585)
HUP		−0.063	0.033	0.044	−0.131	−0.050	0.062	−0.068	−0.083
		(0.129)	(0.105)	(0.092)	(0.125)	(0.116)	(0.085)	(0.118)	(0.114)
Conflict Zone x HUP		−13.422	−11.096	−18.892	8.742	−3.731	−8.629	3.321	−1.206
		(18.309)	(17.120)	(21.895)	(16.297)	(12.878)	(14.074)	(13.365)	(11.176)
Num.Obs.	6,111	6,111	6,111	6,111	6,111	6,111	6,111	6,111	6,111
1st Stage F-Statistic	12.85								

*Note: The sample consists of children and adolescents (aged <19) living in the driest areas of Nigeria (SMI < national mean). All coefficients are calculated using 2SLS and including the following control variables: Age, Female, Rural, Wealth Index, and Religion. Household survey data are taken from UNICEF MICS (2021), conflict data from the UPPSALA Conflict Data Program (UCDP, 2021), rainfall data from the Climate Hazards Group at UCSB (2020-2021), and on soil moisture data from NASA (2020-2021). *p<0.1; **p<0.05; ***p<0.01*

Table 4. *Heterogeneous Effects of Conflict on Child Food Insecurity in Nigeria: Retirement Pension*

	Conflict Zone (1st Stage)	A Day Without Eating (2nd Stage)	Hungry and Did Not Eat (2nd Stage)	Ran Out of Food (2nd Stage)	Skip a Meal (2nd Stage)	Eating Less (2nd Stage)	Ate Few Kinds of Food (2nd Stage)	Not Eating Healthy (2nd Stage)	Worried About Food (2nd Stage)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
IV: Average Rainfall (2021)	−0.002***								
	(0.001)								
Conflict Zone (5Km)		−0.053	0.323	0.237	0.497	0.6	0.561	0.377	0.11
		(0.416)	(0.430)	(0.415)	(0.374)	(0.427)	(0.398)	(0.366)	(0.360)
Pension		0.111	0.09	0.089	0.073	0.094	0.08	0.08	0.097
		(0.080)	(0.083)	(0.077)	(0.078)	(0.077)	(0.078)	(0.078)	(0.079)
Conflict Zone x Pension		−43.730	1.488	0.452	−3.348	−3.075	−6.796	−6.537	−6.296
		(57.287)	(18.940)	(17.681)	(17.477)	(17.537)	(17.575)	(17.093)	(16.999)
Num.Obs.	8,537	8,537	8,537	8,537	8,537	8,537	8,537	8,537	8,537
1st Stage F-Statistic	19.26								

*Note: The sample consists of children and adolescents (aged <19) living in the driest areas of Nigeria (SMI < national mean). All coefficients are calculated using 2SLS and including the following control variables: Age, Female, Rural, Wealth Index, and Religion. Household survey data are taken from UNICEF MICS (2021), conflict data from the UPPSALA Conflict Data Program (UCDP, 2021), rainfall data from the Climate Hazards Group at UCSB (2020-2021), and on soil moisture data from NASA (2020-2021). *p<0.1; **p<0.05; ***p<0.01*

APPENDIX

Appendix Table 1. *Robustness Check: Effect of Conflict on Child Food Insecurity in Nigeria – 10Km Buffer Zone*

	Conflict Zone (1st Stage)	A Day Without Eating (2nd Stage)	Hungry and Did Not Eat (2nd Stage)	Ran Out of Food (2nd Stage)	Skip a Meal (2nd Stage)	Eating Less (2nd Stage)	Ate Few Kinds of Food (2nd Stage)	Not Eating Healthy (2nd Stage)	Worried About Food (2nd Stage)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
IV: Average Rainfall (2021)	−0.004***								
	(0.001)								
Conflict Zone (10Km)		−0.031	0.267	0.214	0.571**	0.467*	0.657**	0.539**	0.294
		(0.277)	(0.271)	(0.259)	(0.246)	(0.261)	(0.266)	(0.252)	(0.240)
Cash Transfers (N-Power)		0.021	−0.020	−0.106	0.182**	0.099	0.163*	0.176**	0.088
		(0.117)	(0.122)	(0.131)	(0.074)	(0.097)	(0.088)	(0.074)	(0.085)
Conflict Zone x Cash Transfers		−0.962	−1.089	−0.778	−2.278**	−1.939**	−2.296**	−2.180**	−1.874**
		(0.793)	(0.804)	(0.789)	(0.959)	(0.903)	(0.951)	(0.933)	(0.914)
Num.Obs.	9,168	9,168	9,168	9,168	9,168	9,168	9,168	9,168	9,168
1st Stage F-Statistic	38.17								

*Note: The sample consists of children and adolescents (aged <19) living in the driest areas of Nigeria (SMI < national mean). All coefficients are calculated using 2SLS and including the following control variables: Age, Female, Rural, Wealth Index, and Religion. Household survey data are taken from UNICEF MICS (2021), conflict data from the UPPSALA Conflict Data Program (UCDP, 2021), rainfall data from the Climate Hazards Group at UCSB (2020-2021), and on soil moisture data from NASA (2020-2021). *p<0.1; **p<0.05; ***p<0.01*

Appendix Table 2. *Robustness Check: Effect of Conflict on Child Food Insecurity in Nigeria – 15Km Buffer Zone*

	Conflict Zone (1st Stage)	A Day Without Eating (2nd Stage)	Hungry and Did Not Eat (2nd Stage)	Ran Out of Food (2nd Stage)	Skip a Meal (2nd Stage)	Eating Less (2nd Stage)	Ate Few Kinds of Food (2nd Stage)	Not Eating Healthy (2nd Stage)	Worried About Food (2nd Stage)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
IV: Average Rainfall (2021)	−0.004***								
	(0.001)								
Conflict Zone (15Km)		−0.027	0.255	0.204	0.545**	0.446*	0.626**	0.515**	0.282
		(0.262)	(0.257)	(0.245)	(0.231)	(0.247)	(0.251)	(0.238)	(0.227)
Cash Transfers (N-Power)		0.042	−0.002	−0.093	0.221***	0.133	0.201**	0.213***	0.123
		(0.127)	(0.133)	(0.142)	(0.082)	(0.105)	(0.096)	(0.081)	(0.092)
Conflict Zone x Cash Transfers		−1.001	−1.096	−0.780	−2.291**	−1.953**	−2.300**	−2.194**	−1.907**
		(0.824)	(0.828)	(0.809)	(0.992)	(0.929)	(0.978)	(0.965)	(0.946)
Num.Obs.	9,168	9,168	9,168	9,168	9,168	9,168	9,168	9,168	9,168
1st Stage F-Statistic	40.76								

*Note: The sample consists of children and adolescents (aged <19) living in the driest areas of Nigeria (SMI < national mean). All coefficients are calculated using 2SLS and including the following control variables: Age, Female, Rural, Wealth Index, and Religion. Household survey data are taken from UNICEF MICS (2021), conflict data from the UPPSALA Conflict Data Program (UCDP, 2021), rainfall data from the Climate Hazards Group at UCSB (2020-2021), and on soil moisture data from NASA (2020-2021). *p<0.1; **p<0.05; ***p<0.01*

Appendix Table 3. *Robustness Check: Effect of Conflict on Child Food Insecurity in Nigeria – 20Km Buffer Zone*

	Conflict Zone (1st Stage)	A Day Without Eating (2nd Stage)	Hungry and Did Not Eat (2nd Stage)	Ran Out of Food (2nd Stage)	Skip a Meal (2nd Stage)	Eating Less (2nd Stage)	Ate Few Kinds of Food (2nd Stage)	Not Eating Healthy (2nd Stage)	Worried About Food (2nd Stage)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
IV: Average Rainfall (2021)	−0.004***								
	(0.001)								
Conflict Zone (20Km)		−0.028	0.268	0.214	0.573**	0.468*	0.658**	0.541**	0.297
		(0.276)	(0.270)	(0.258)	(0.248)	(0.263)	(0.268)	(0.253)	(0.239)
Cash Transfers (N-Power)		0.041	0.004	−0.089	0.233***	0.142	0.214**	0.224***	0.129
		(0.127)	(0.135)	(0.144)	(0.084)	(0.107)	(0.099)	(0.084)	(0.094)
Conflict Zone x Cash Transfers		−0.999	−1.108	−0.790	−2.316**	−1.973**	−2.329**	−2.218**	−1.920**
		(0.826)	(0.831)	(0.812)	(0.998)	(0.935)	(0.985)	(0.971)	(0.950)
Num.Obs.	9,168	9,168	9,168	9,168	9,168	9,168	9,168	9,168	9,168
1st Stage F-Statistic	32.68								

*Note: The sample consists of children and adolescents (aged <19) living in the driest areas of Nigeria (SMI < national mean). All coefficients are calculated using 2SLS and including the following control variables: Age, Female, Rural, Wealth Index, and Religion. Household survey data are taken from UNICEF MICS (2021), conflict data from the UPPSALA Conflict Data Program (UCDP, 2021), rainfall data from the Climate Hazards Group at UCSB (2020-2021), and on soil moisture data from NASA (2020-2021). *p<0.1; **p<0.05; ***p<0.01*

Appendix Table 4. *Robustness Check: Effect of Conflict on Child Food Insecurity in Nigeria – 25Km Buffer Zone*

	Conflict Zone (1st Stage)	A Day Without Eating (2nd Stage)	Hungry and Did Not Eat (2nd Stage)	Ran Out of Food (2nd Stage)	Skip a Meal (2nd Stage)	Eating Less (2nd Stage)	Ate Few Kinds of Food (2nd Stage)	Not Eating Healthy (2nd Stage)	Worried About Food (2nd Stage)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
IV: Average Rainfall (2021)	−0.003***								
	(0.001)								
Conflict Zone (25Km)		−0.027	0.254	0.203	0.543**	0.444*	0.624**	0.513**	0.281
		(0.262)	(0.256)	(0.244)	(0.234)	(0.248)	(0.253)	(0.239)	(0.226)
Cash Transfers (N-Power)		0.041	0.004	−0.088	0.233***	0.143	0.215**	0.225***	0.129
		(0.128)	(0.134)	(0.144)	(0.084)	(0.107)	(0.098)	(0.084)	(0.093)
Conflict Zone x Cash Transfers		−1.001	−1.091	−0.776	−2.280**	−1.943**	−2.287**	−2.183**	−1.901**
		(0.823)	(0.826)	(0.807)	(0.988)	(0.925)	(0.973)	(0.961)	(0.943)
Num.Obs.	9,168	9,168	9,168	9,168	9,168	9,168	9,168	9,168	9,168
1st Stage F-Statistic	33.5								

*Note: The sample consists of children and adolescents (aged <19) living in the driest areas of Nigeria (SMI < national mean). All coefficients are calculated using 2SLS and including the following control variables: Age, Female, Rural, Wealth Index, and Religion. Household survey data are taken from UNICEF MICS (2021), conflict data from the UPPSALA Conflict Data Program (UCDP, 2021), rainfall data from the Climate Hazards Group at UCSB (2020-2021), and on soil moisture data from NASA (2020-2021). *p<0.1; **p<0.05; ***p<0.01*

Appendix Table 5. Placebo Test – Effect of Conflict on Food Insecurity Using Different Rainfall Years: Lag 1 Year

	Conflict Zone (1st Stage)	A Day Without Eating (2nd Stage)	Hungry and Did Not Eat (2nd Stage)	Ran Out of Food (2nd Stage)	Skip a Meal (2nd Stage)	Eating Less (2nd Stage)	Ate Few Kinds of Food (2nd Stage)	Not Eating Healthy (2nd Stage)	Worried About Food (2nd Stage)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
IV: Average Nearby Rainfall (5Km)	−0.001***								
	(0.000)								
Conflict Zone (5Km)		−0.922	−0.653	−0.796	0.093	0.132	0.308	−0.107	−0.454
		(0.606)	(0.581)	(0.516)	(0.462)	(0.487)	(0.477)	(0.443)	(0.442)
Cash Transfers (N-Power)		0.027	0.016	−0.113	0.157	0.051	0.17	0.153	0.028
		(0.143)	(0.131)	(0.154)	(0.130)	(0.150)	(0.142)	(0.129)	(0.138)
Conflict Zone x Cash Transfers		−0.883	−1.519	−0.614	−2.298	−1.640	−2.771	−2.161	−1.245
		(1.551)	(1.740)	(1.454)	(1.798)	(1.524)	(2.010)	(1.816)	(1.534)
Num.Obs.	9,168	9,168	9,168	9,168	9,168	9,168	9,168	9,168	9,168
1st Stage F-Statistic	16.96								

*Note: The sample consists of children and adolescents (aged <19) living in the driest areas of Nigeria (SMI < national mean). All coefficients are calculated using 2SLS and including the following control variables: Age, Female, Rural, Wealth Index, and Religion. Household survey data are taken from UNICEF MICS (2021), conflict data from the UPPSALA Conflict Data Program (UCDP, 2021), rainfall data from the Climate Hazards Group at UCSB (2020-2021), and on soil moisture data from NASA (2020-2021). *p<0.1; **p<0.05; ***p<0.01*

Appendix Table 6. Placebo Test – Effect of Conflict on Food Insecurity Using Different Rainfall Years: Lag 2 Year

	Conflict Zone (1st Stage)	A Day Without Eating (2nd Stage)	Hungry and Did Not Eat (2nd Stage)	Ran Out of Food (2nd Stage)	Skip a Meal (2nd Stage)	Eating Less (2nd Stage)	Ate Few Kinds of Food (2nd Stage)	Not Eating Healthy (2nd Stage)	Worried About Food (2nd Stage)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
IV: Average Nearby Rainfall (5Km)	−0.001								
	(0.000)								
Conflict Zone (5Km)		−1.239	−2.750	−3.258	−1.561	−1.860	−2.323	−2.476	−1.239
		(1.599)	(2.413)	(2.503)	(1.650)	(1.830)	(2.015)	(2.093)	(1.599)
Cash Transfers (N-Power)		0.092	−0.029	−0.169	0.169	0.076	0.101	0.126	0.092
		(0.171)	(0.249)	(0.260)	(0.169)	(0.198)	(0.202)	(0.199)	(0.171)
Conflict Zone x Cash Transfers		−1.845	−0.493	0.619	−2.055	−1.477	−1.336	−1.297	−1.845
		(2.769)	(3.451)	(3.305)	(3.035)	(3.028)	(3.220)	(3.305)	(2.769)
Num.Obs.	9,168	9,168	9,168	9,168	9,168	9,168	9,168	9,168	9,168
1st Stage F-Statistic	2.23								

*Note: The sample consists of children and adolescents (aged <19) living in the driest areas of Nigeria (SMI < national mean). All coefficients are calculated using 2SLS and including the following control variables: Age, Female, Rural, Wealth Index, and Religion. Household survey data are taken from UNICEF MICS (2021), conflict data from the UPPSALA Conflict Data Program (UCDP, 2021), rainfall data from the Climate Hazards Group at UCSB (2020-2021), and on soil moisture data from NASA (2020-2021). *p<0.1; **p<0.05; ***p<0.01*

Appendix Table 7. Placebo Test: Effect of Conflict on Food Insecurity Using Nearby Rainfall

	Conflict Zone (1st Stage)	A Day Without Eating (2nd Stage)	Hungry and Did Not Eat (2nd Stage)	Ran Out of Food (2nd Stage)	Skip a Meal (2nd Stage)	Eating Less (2nd Stage)	Ate Few Kinds of Food (2nd Stage)	Not Eating Healthy (2nd Stage)	Worried About Food (2nd Stage)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
IV: Average Nearby Rainfall (5Km)	0.000***								
	(0.001)								
Conflict Zone (5Km)		0.86	0.497	0.816	0.415	0.49	0.534	0.896	0.958
		(0.893)	(0.817)	(0.820)	(0.773)	(0.782)	(0.781)	(0.771)	(0.788)
Cash Transfers (N-Power)		−0.069	−0.046	0.094	0.188	0.103	0.007	0.2	0.247
		(0.334)	(0.328)	(0.346)	(0.223)	(0.270)	(0.260)	(0.222)	(0.243)
Conflict Zone x Cash Transfers		−0.140	−1.028	−3.490	−2.739	−2.348	−0.856	−2.963	−4.214
		(4.158)	(3.915)	(4.570)	(3.227)	(3.606)	(3.511)	(3.138)	(3.841)
Num.Obs.	9,168	9,168	9,168	9,168	9,168	9,168	9,168	9,168	9,168
1st Stage F-Statistic	7.86								

*Note: The sample consists of children and adolescents (aged <19) living in the driest areas of Nigeria (SMI < national mean). All coefficients are calculated using 2SLS and including the following control variables: Age, Female, Rural, Wealth Index, and Religion. Household survey data are taken from UNICEF MICS (2021), conflict data from the UPPSALA Conflict Data Program (UCDP, 2021), rainfall data from the Climate Hazards Group at UCSB (2020-2021), and on soil moisture data from NASA (2020-2021). *p<0.1; **p<0.05; ***p<0.01*